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BIOM1051

Module 1: Principles of Cell Function

Lecture 3

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Module 1: Principles of Cell Function

- Membrane Structure & Function – Lecture 1 & 2
- Cell Communication & Receptor Families – Lecture 3
- Review & Application – Workshop 2
- + the Osmosis Practical

Cell Communication & Receptor Families: Learning Objectives – Lectures 3

- Describe the types of cell to cell communication
- List and describe in general terms the 3 stages of cell signalling
- Briefly compare and contrast the 4 families of receptors
- Explain the concept of signal amplification
- Describe how G protein-coupled receptors produce a response in a cell
- Rationalise the effects of insulin and steroids at the receptor and cellular level

Cell Communication & Receptor Families: Resources

- Campbell Biology, 12th ed, 2021 – Chapter 11
  - Concept 11.1: External signals are converted to responses within the cell
  - Concept 11.2: Reception: A signalling molecule binds to a receptor protein, causing it to change shape
  - Concept 11.3: Transduction: Cascades of molecular interactions relay signals from receptors to target molecules in the cell
  - Concept 11.4: Response: Cell signalling leads to regulation of transcription or cytoplasmic activities
- GPCR Mechanism Video: Blackboard
- Mastering Biology

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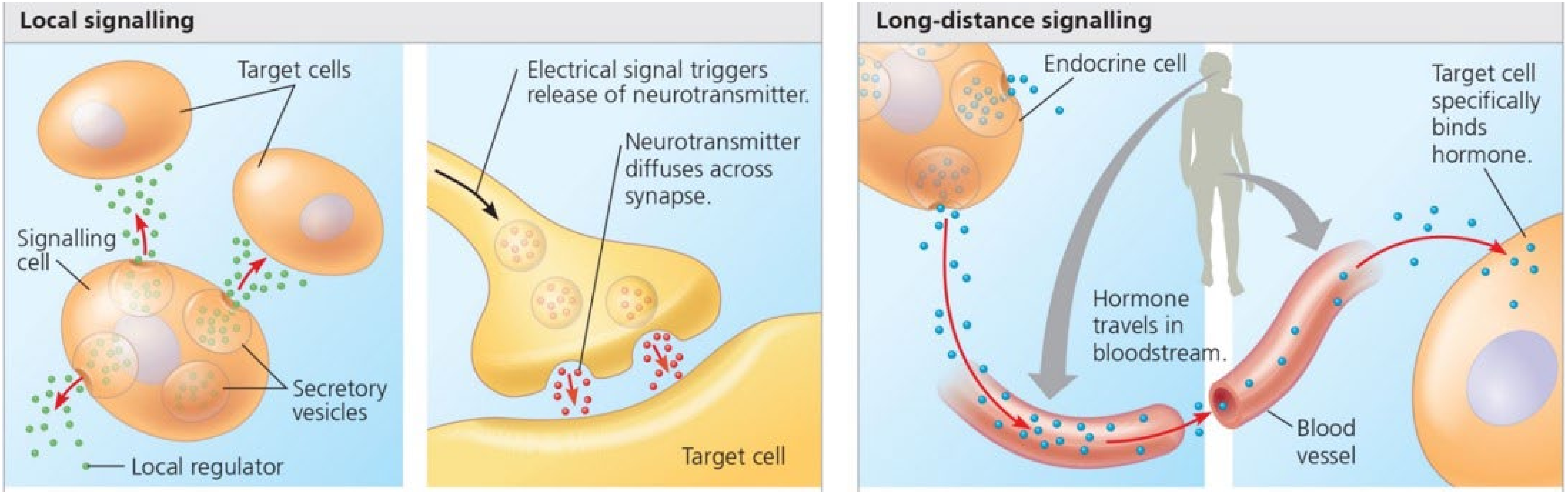
Cell to cell communication:
Local vs. long-distance signalling



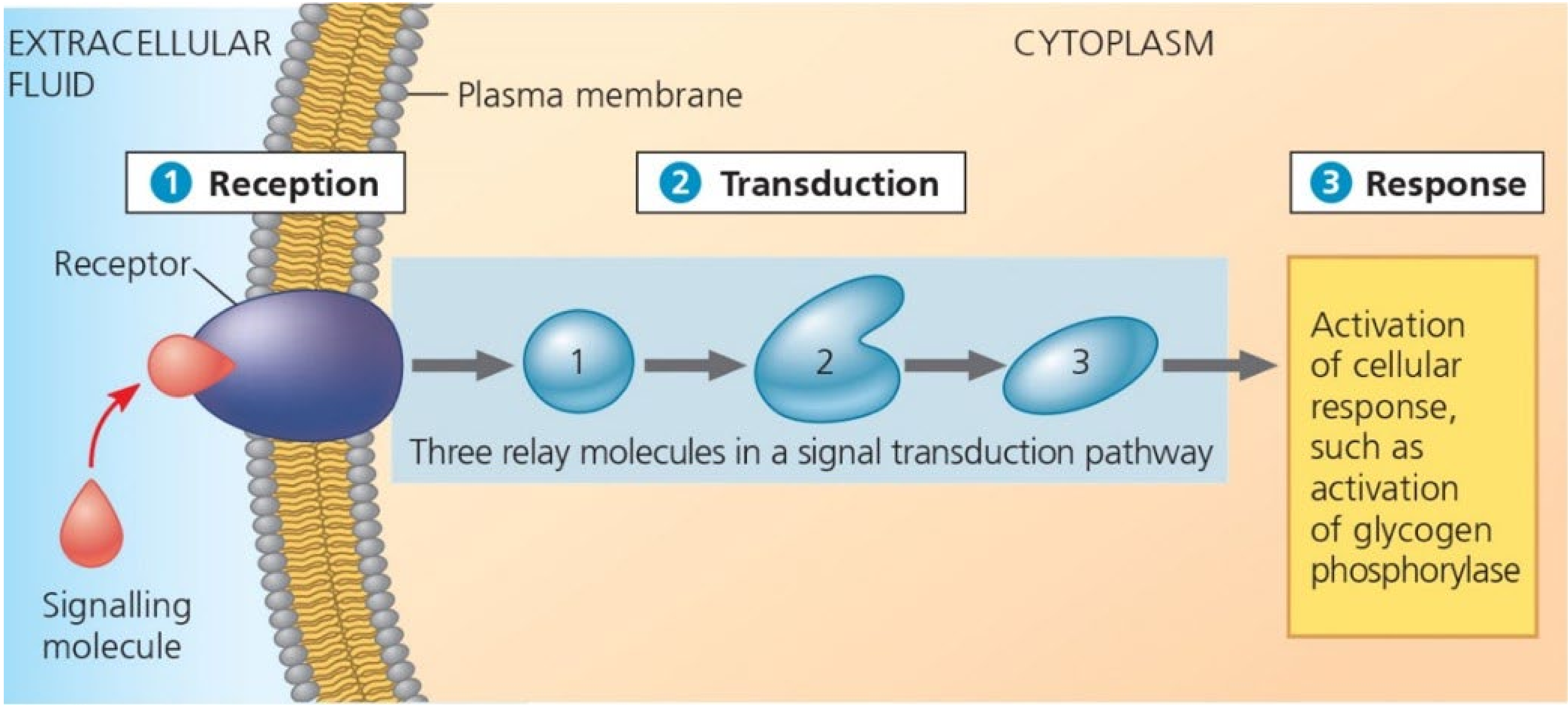
(a) Paracrine signalling. A signalling cell acts on nearby target cells by secreting molecules of a local regulator (a growth factor, for example).

(b) Synaptic signalling. A nerve cell releases neurotransmitter molecules into a synapse, stimulating the target cell, such as a muscle or another nerve cell.

(c) Endocrine (hormonal) signalling. Specialised endocrine cells secrete hormones into body fluids, often blood. Hormones reach most body cells, but are bound by and affect only some cells.

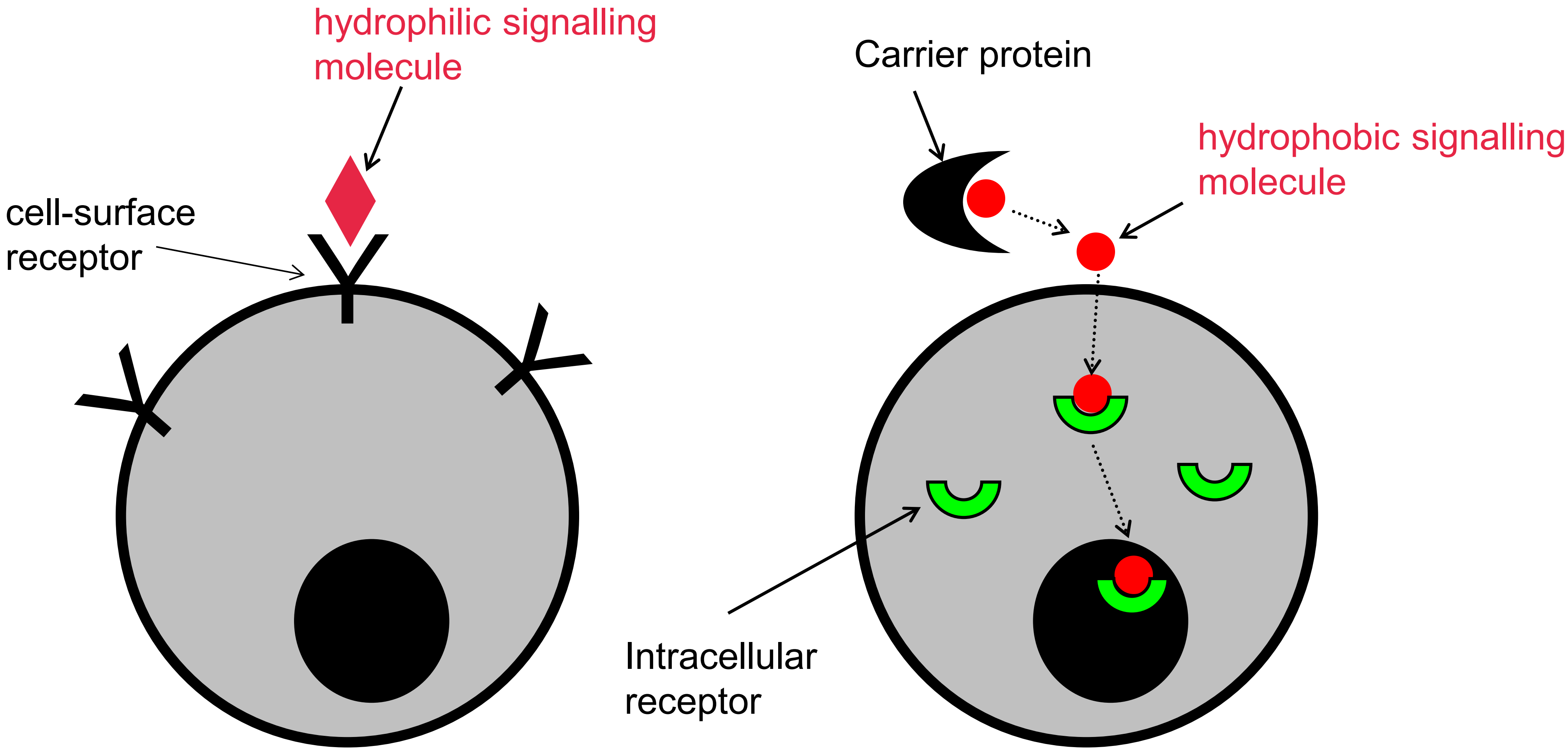
Campbell Biology, 12th ed, Fig 11.5

Cell to cell communication: 3 stages of cell signalling



Campbell Biology, 12th ed, Fig 11.6

Hormone/Ligand/Agonist act via receptors



Response – e.g. gene activation, enzyme stimulation,rearrange cytoskeleton

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Receptor Families

Plasma Membrane Receptors

❖ Ion channel receptors:

- Na^+ channel opened by ligand – e.g. nicotinic receptors for acetylcholine – fast neurotransmission

❖ G protein-coupled receptors: 7 TM-spanning regions

- all aspects of physiology and pharmacology

❖ Receptor tyrosine kinases: e.g. insulin receptors

- metabolism, cell growth, cell reproduction

Intracellular receptors

❖ Steroid receptors

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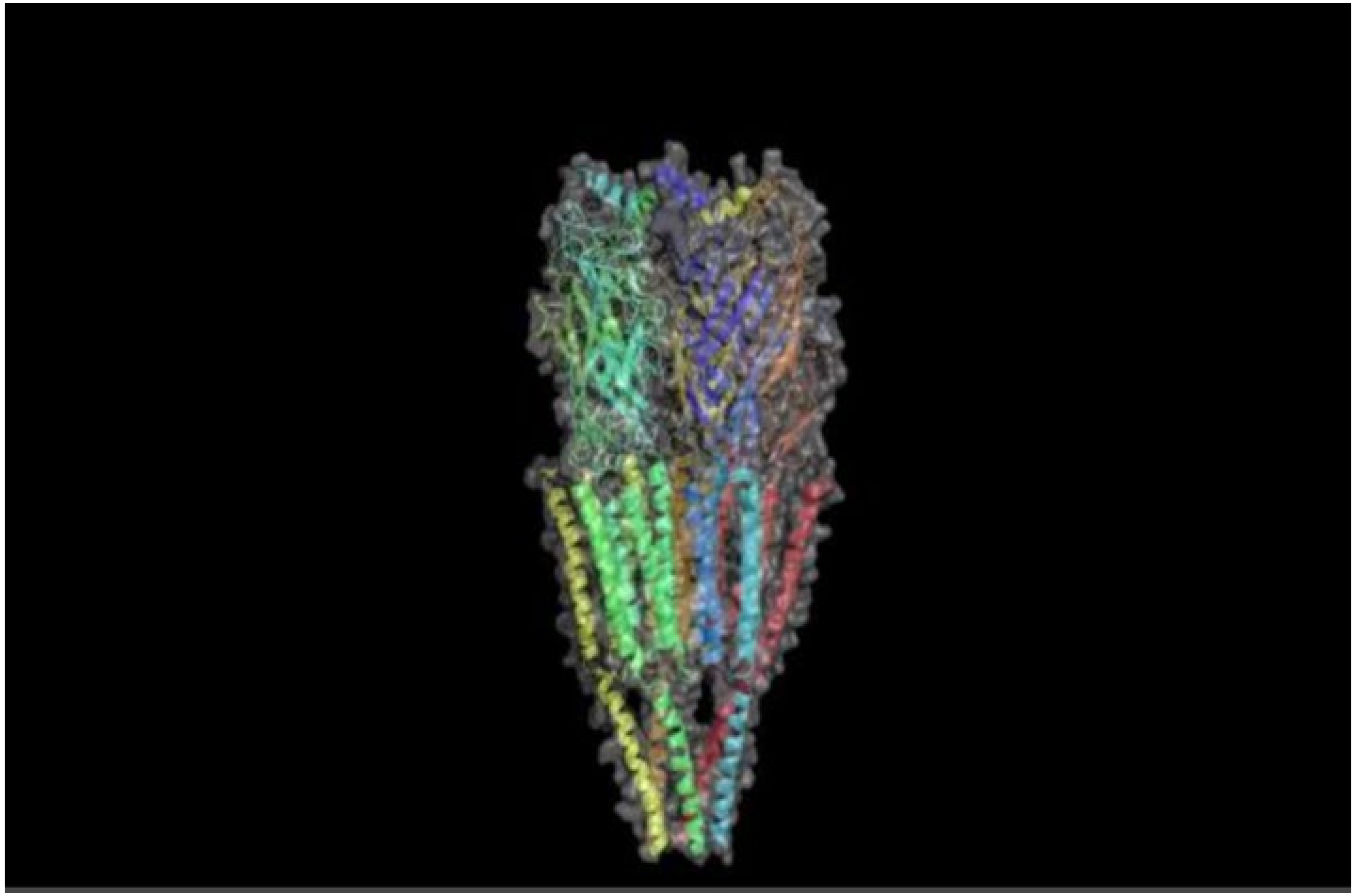
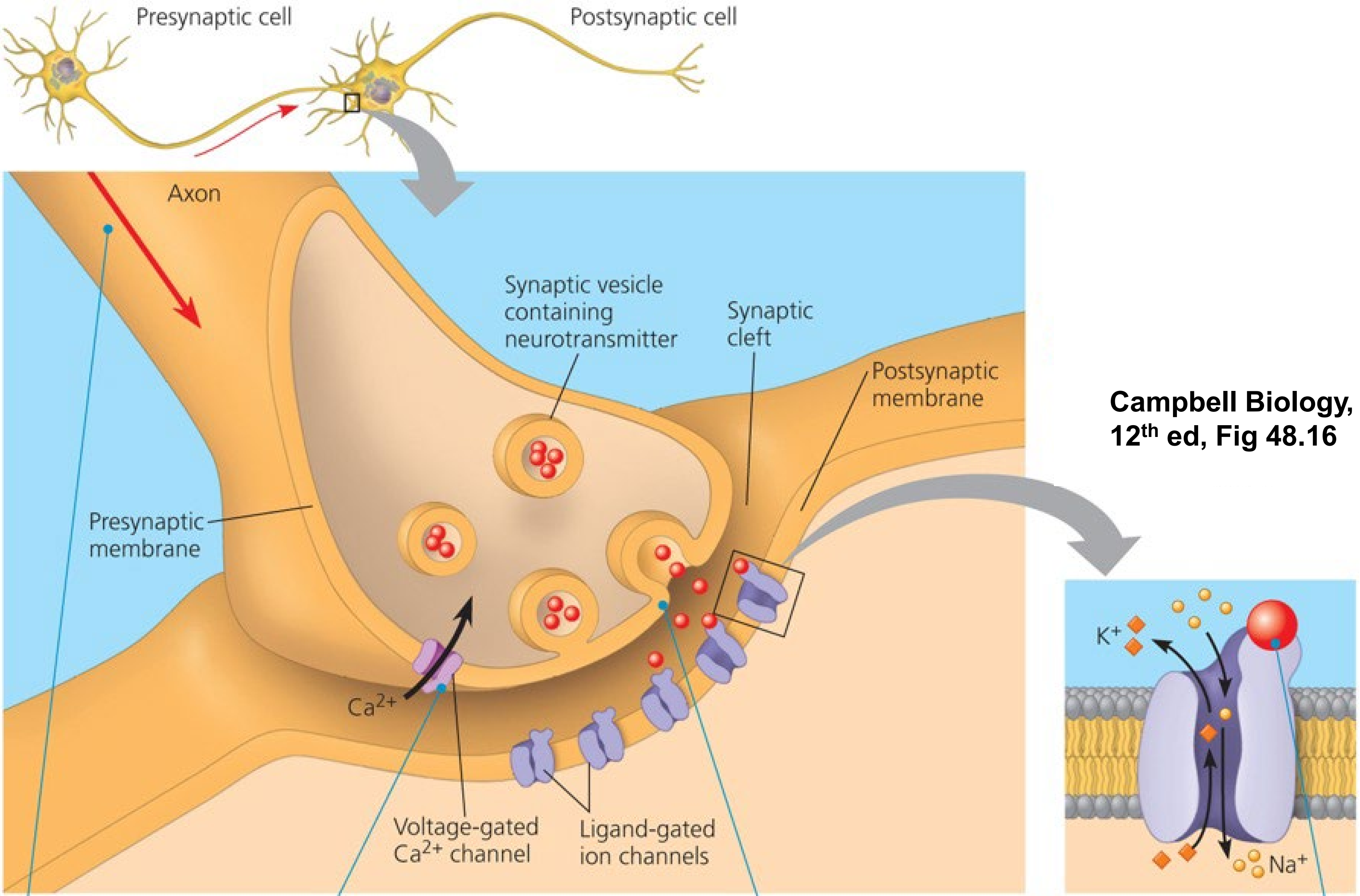
Ion channel receptors

- ligand-gated ion channel
- e.g. nicotinic acetylcholine receptor



Mastering Biology, Chapter 11, Animation

Ion channel linked receptors = Ligand-gated ion channels



Campbell Biology, 12th ed, Fig 48.16

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Intracellular receptors

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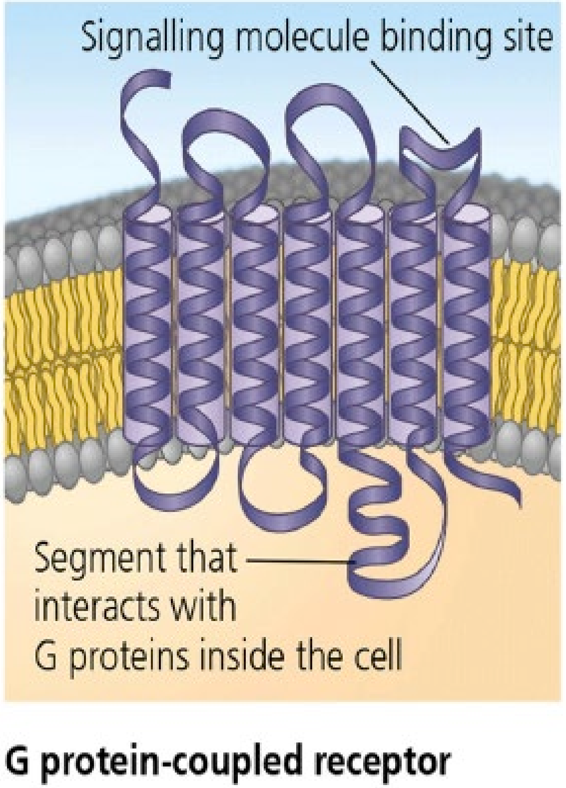
G protein-coupled receptors (GPCRs)

- seven transmembrane spanning domains (7 TMDs)

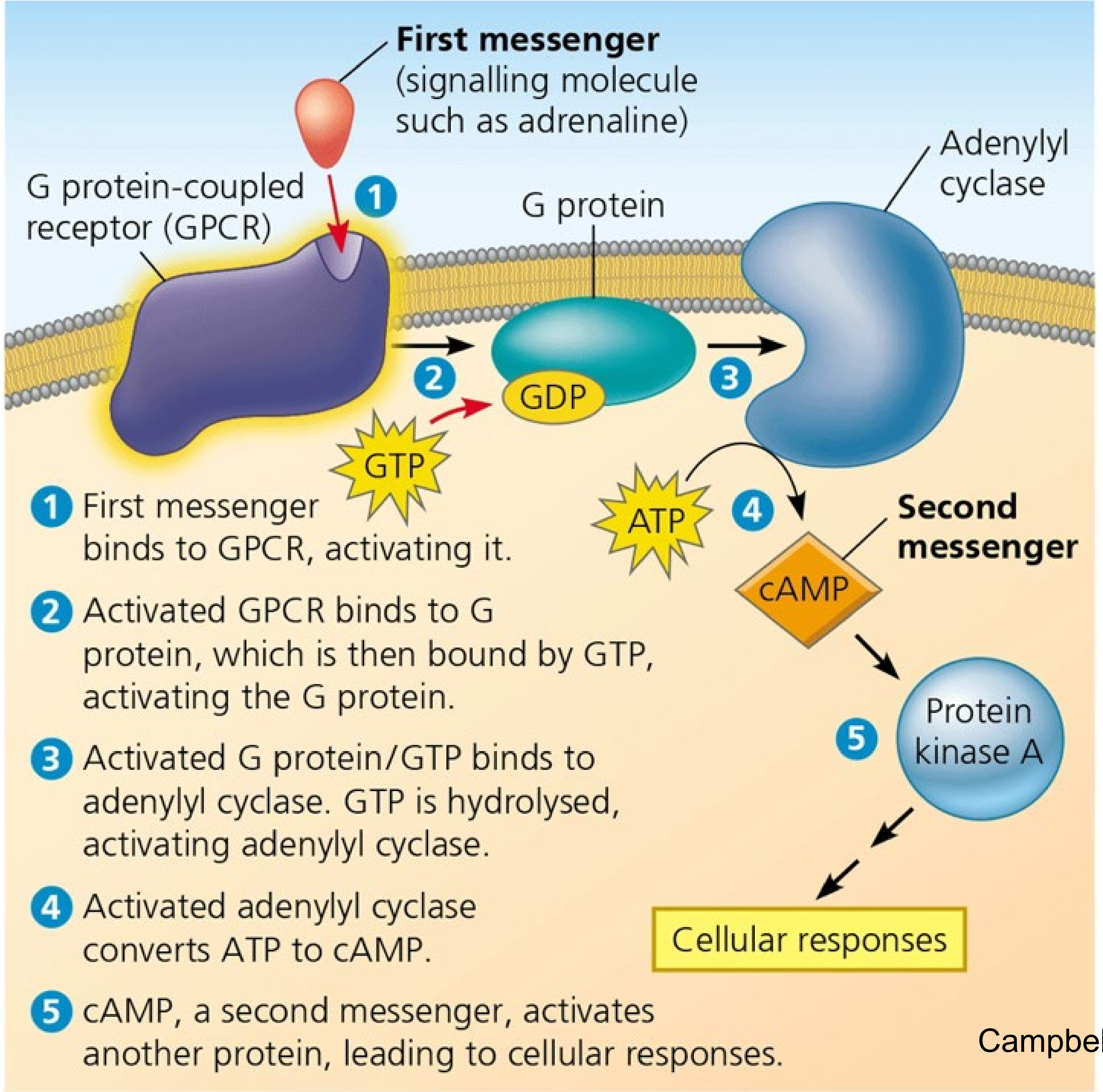
- the largest family of receptors:
  - >1000 members in human genome
  - 150 orphans
  - >50% of current drugs target GPCRs

- activated by a variety of stimuli:
  - light, ions (eg. Ca^{2+}), odourants, gustative molecules, neurotransmitters, hormones, peptides, proteins.

- GPCRs interact with heterotrimeric G proteins to control the activity of enzymes, ion channels, and intracellular signal transduction pathways

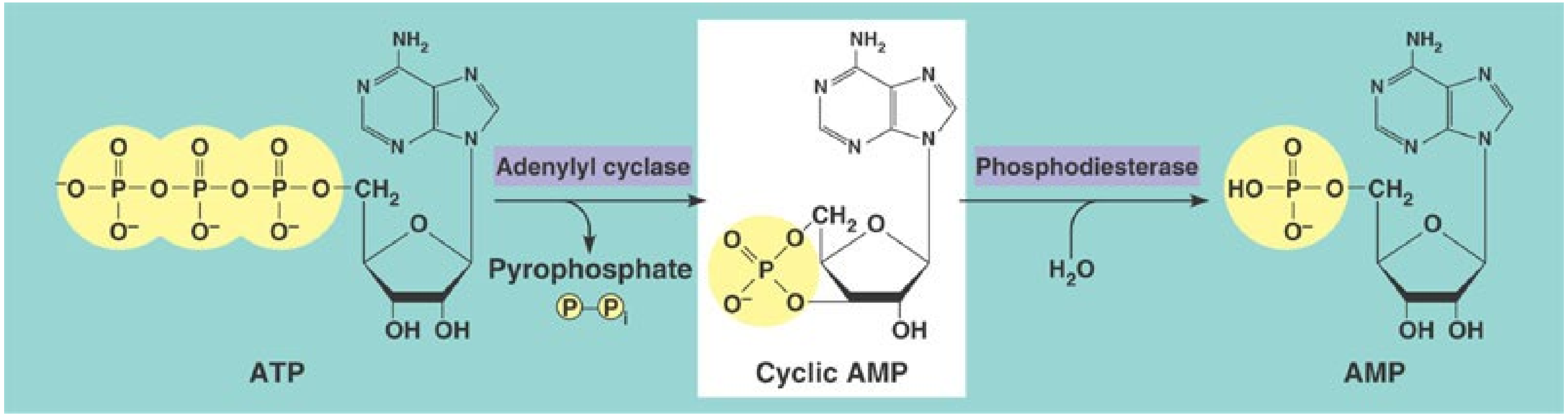


Campbell Biology, 12th ed, Fig. 11.8



Campbell Biology, 12th ed, Fig. 11.12

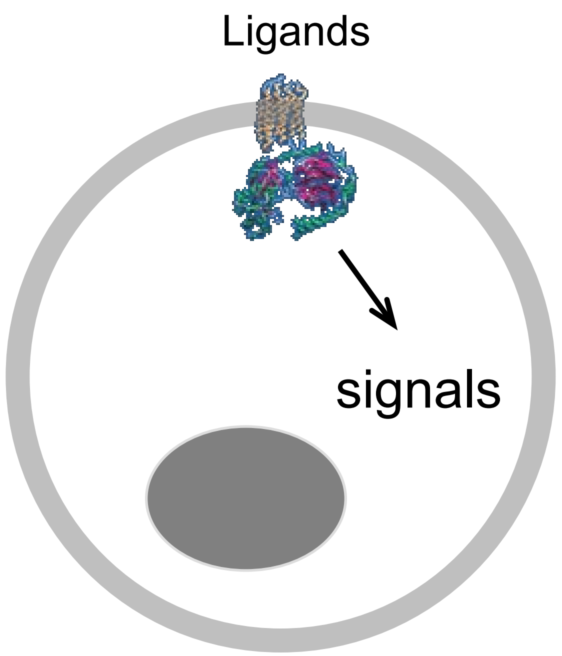
Adenylyl cyclase turns ATP into cyclic AMP

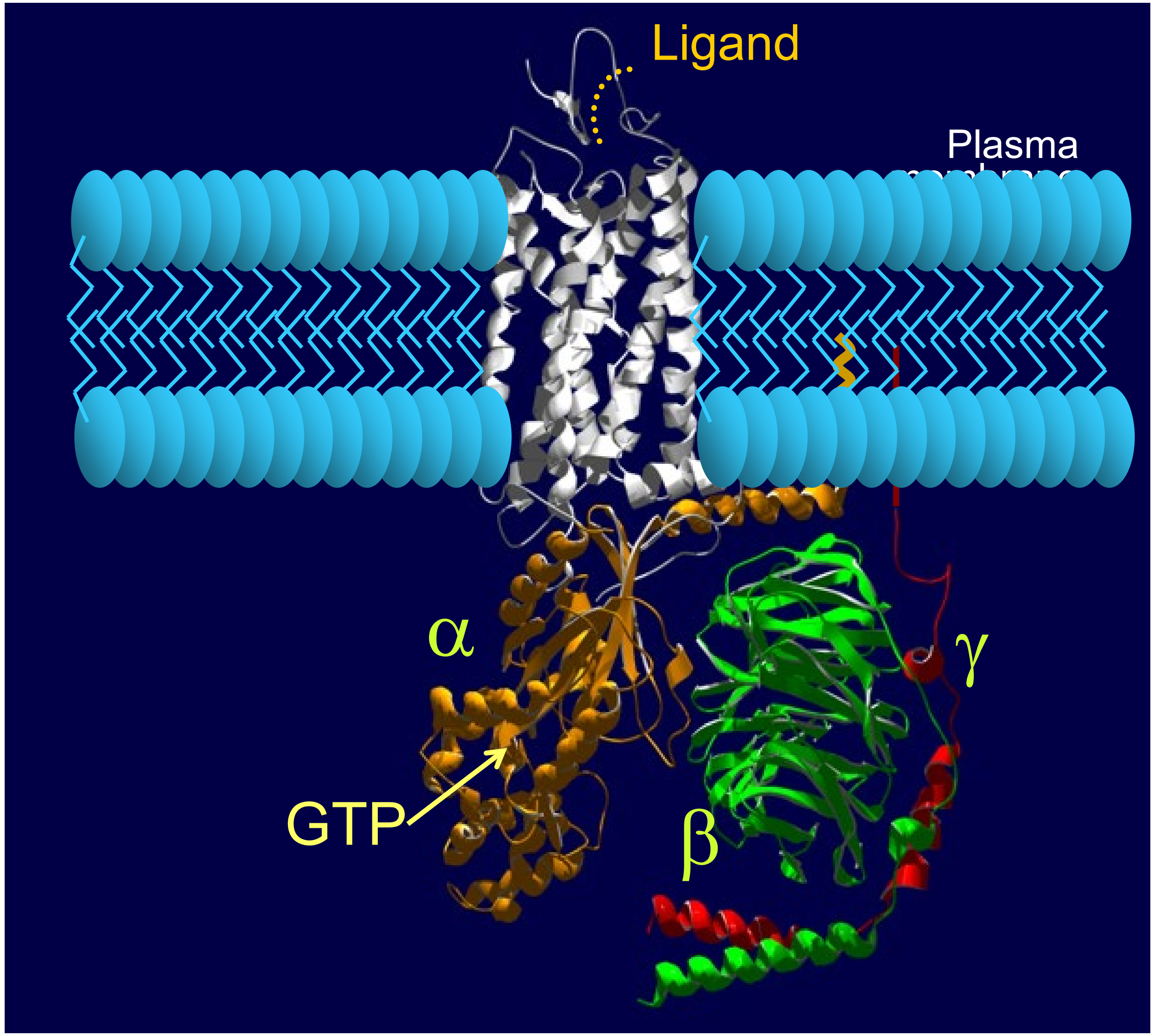


ATP – adenosine triphosphate is also important as a source of cellular energy!!

Campbell Biology, 12th ed, Fig. 11.11 (Note: above version shows more detail)

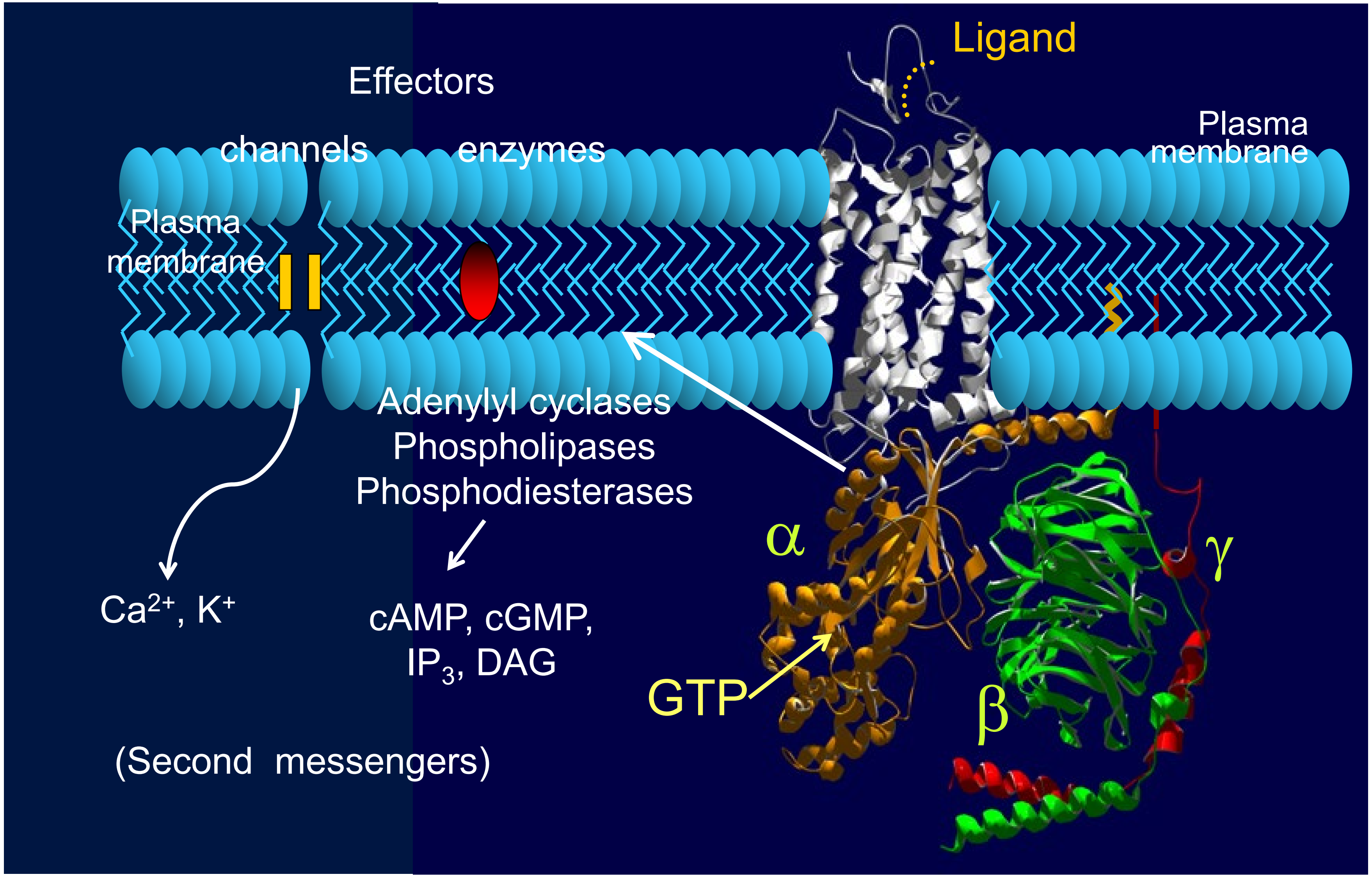
G protein-coupled receptors (GPCRs)





Also see Campbell Biology, 12th ed, Fig 11.7

G protein-coupled receptors (GPCRs)



G-Protein Coupled Receptor (GPCR)

Mastering Biology, Chapter 11, Bioflix

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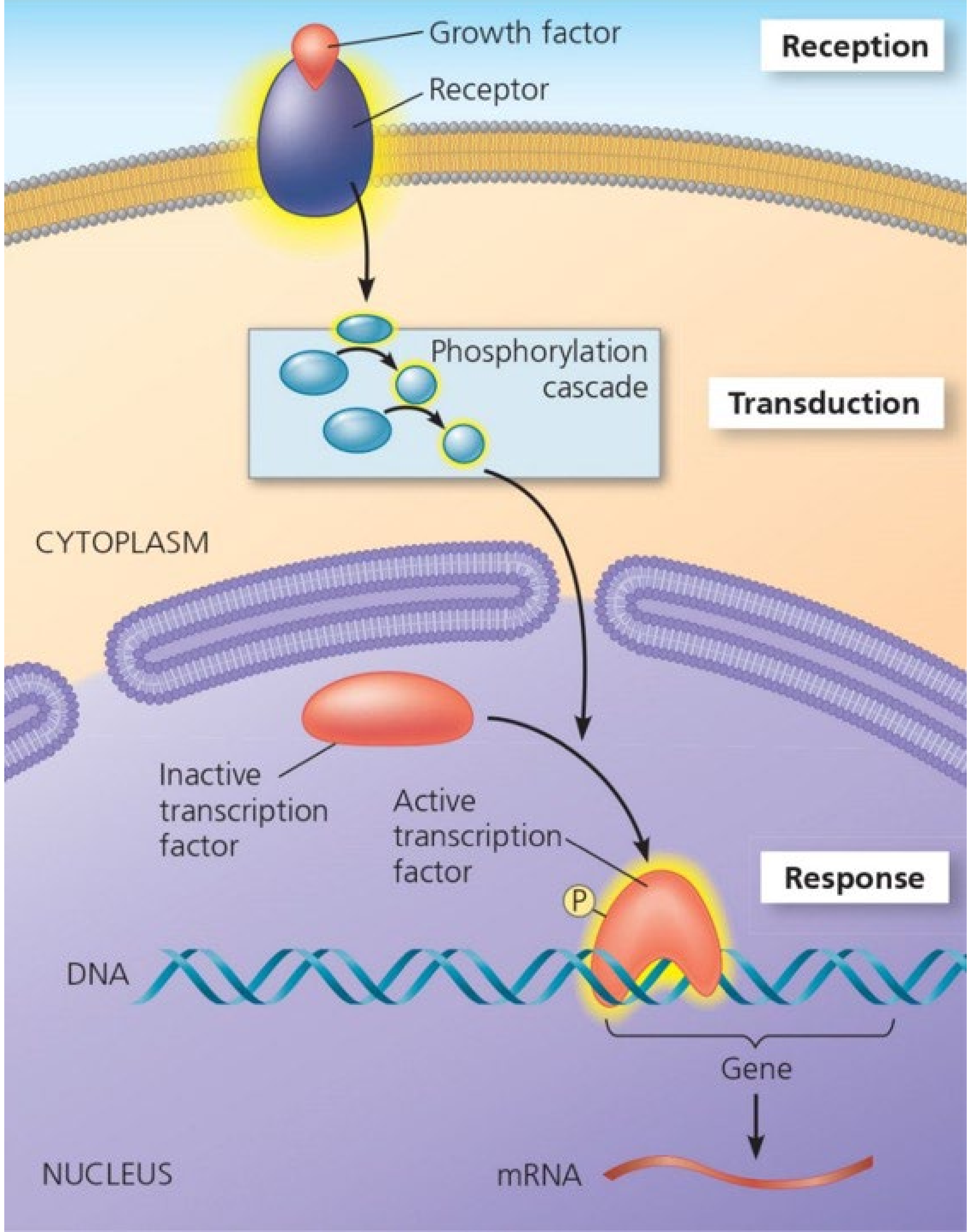
❖ Receptor tyrosine kinases: e.g. insulin receptors

- metabolism, cell growth, cell reproduction

Intracellular receptors

❖ Steroid receptors



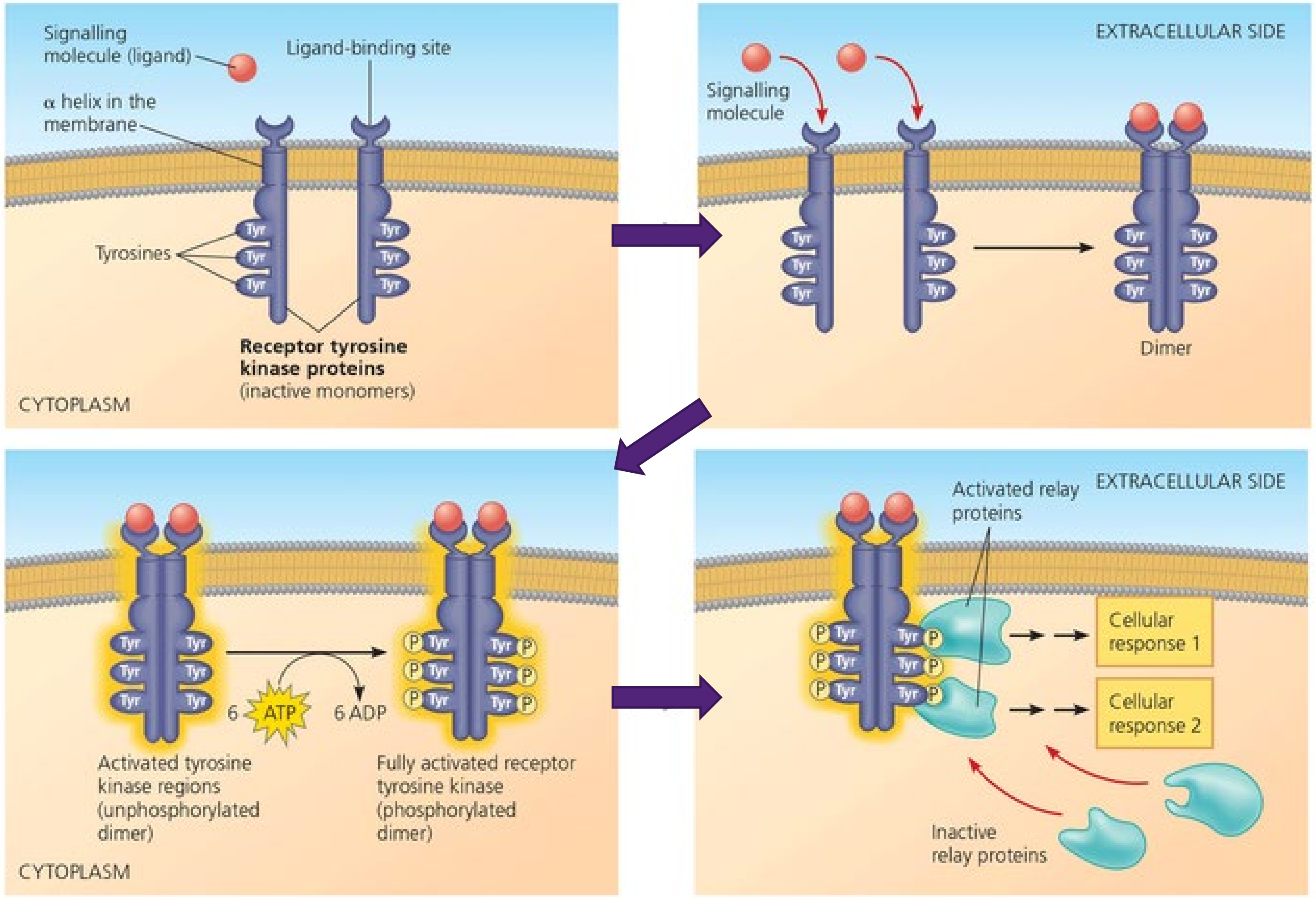
Phosphorylation Cascades…

Mediated by enzymes called KINASES – these transfer phosphate and activate or inactivate target proteins

Phosphate = PO_4^-

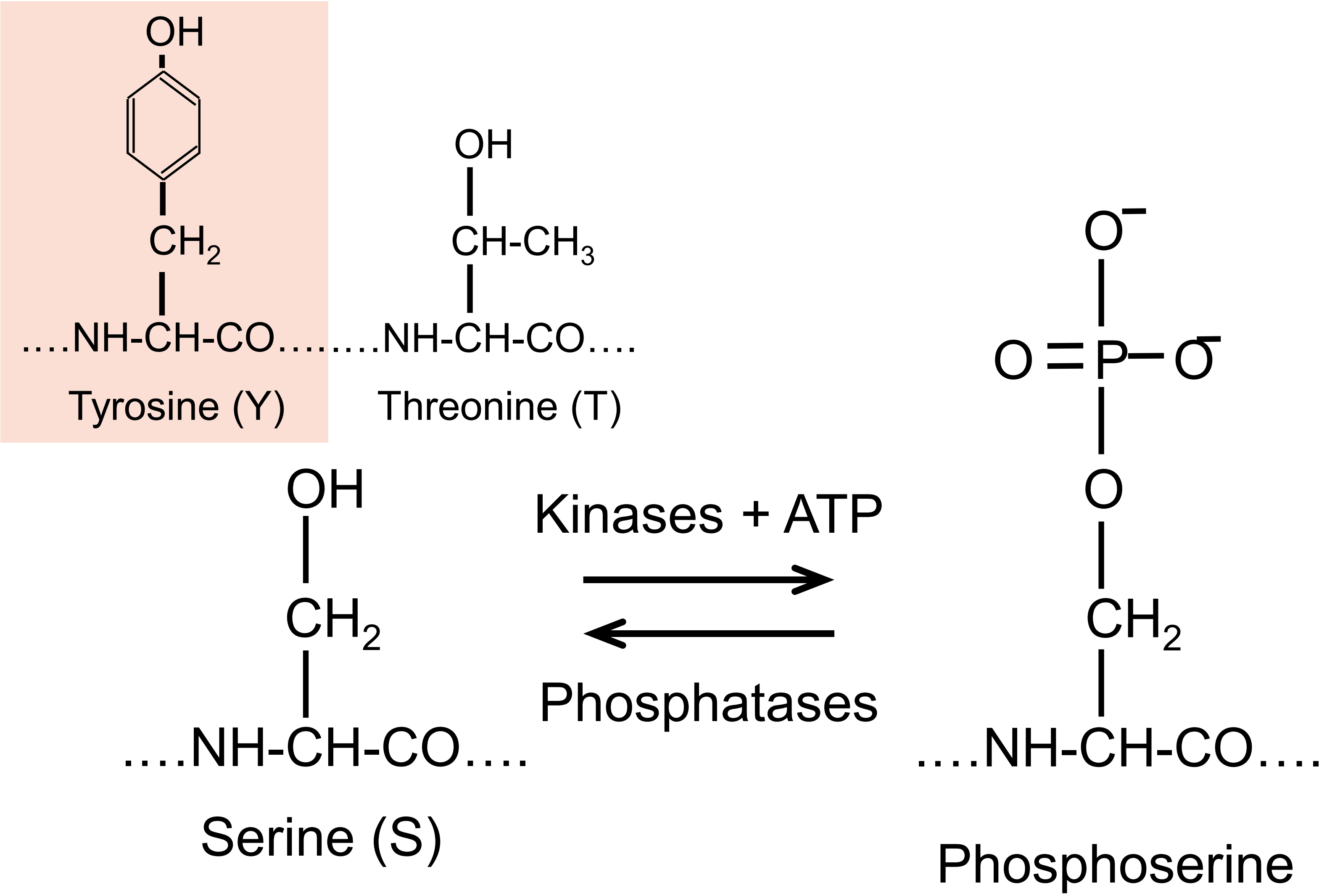
Campbell Biology, 12th ed, Fig. 11.15

Receptor tyrosine kinases (RTKs)

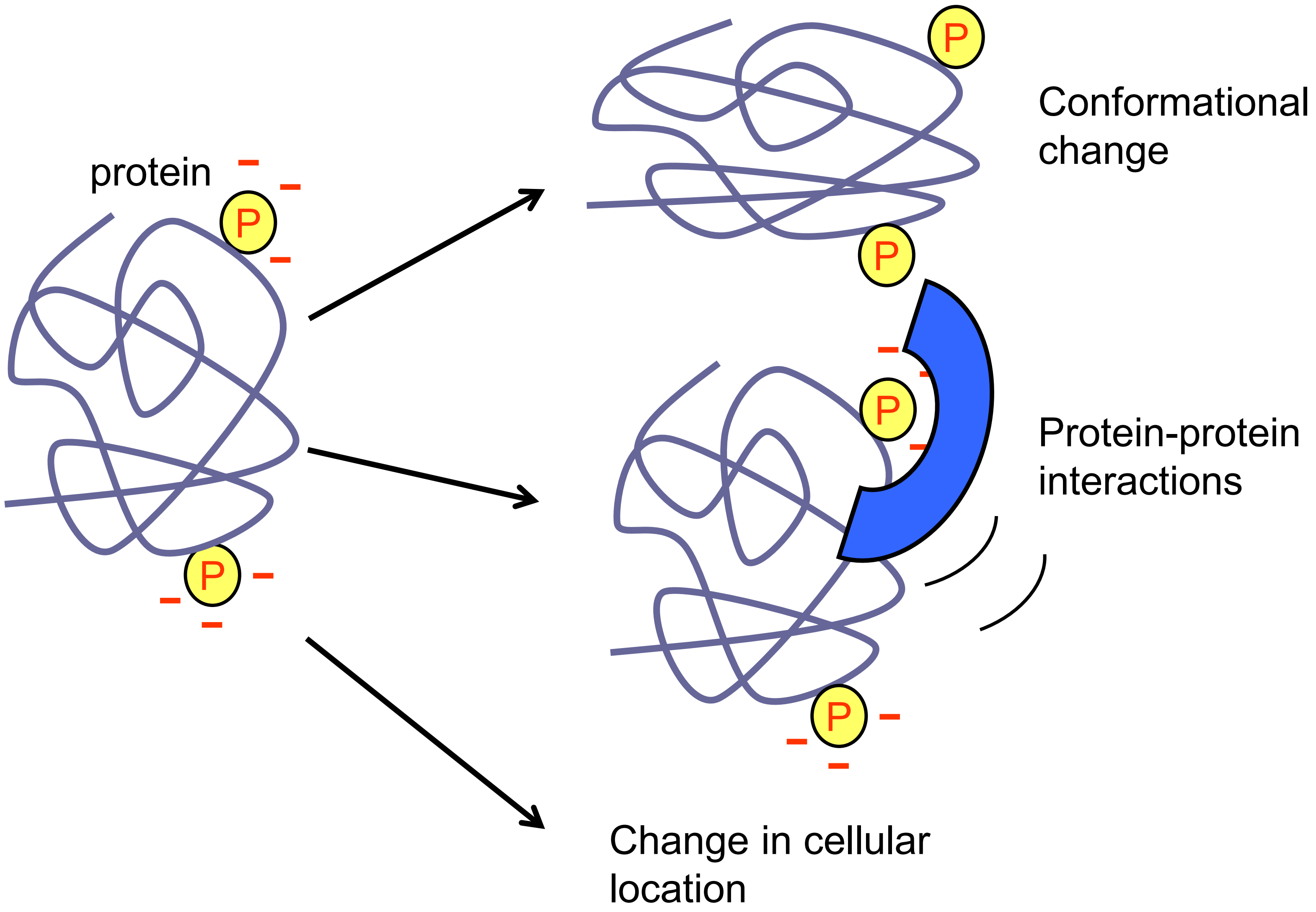


Campbell Biology, 12th ed, Fig. 11.8

Protein phosphorylation



Role of protein phosphorylation



These changes lead to activation or inactivation of the target proteins!

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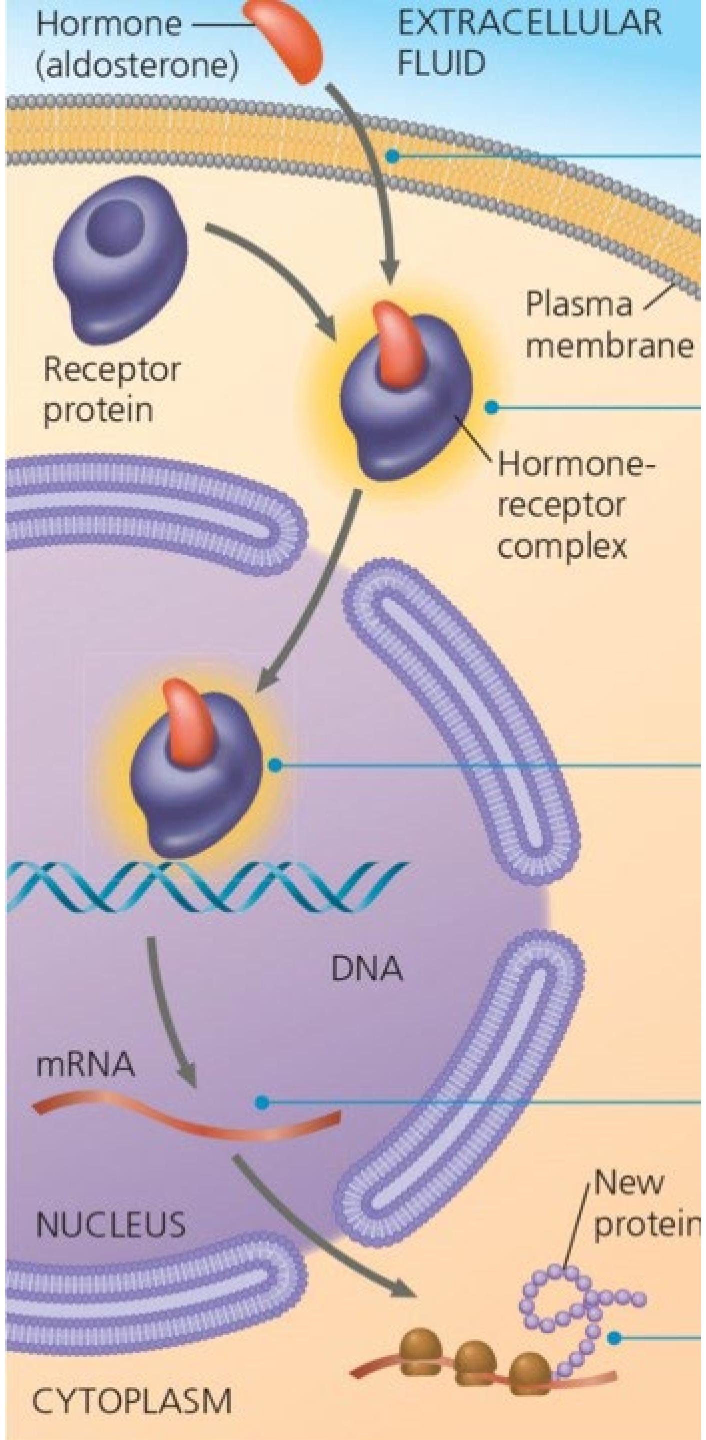
Steroid hormone interacting with an intracellular steroid receptor



Campbell Biology, 12th ed, Fig. 11.9